

Title: Mitochondrial DNA at the interface between mitochondrial dysfunction and neuroinflammation in glaucoma

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Background: Glaucoma is a leading cause of irreversible vision loss characterized by retinal ganglion cell (RGC) degeneration. Mitochondrial DNA (MtDNA) mutations are known to accumulate with age and are linked to neurodegenerative disorders. The retina is known to be a metabolically active tissue and is prone to mitochondrial dysfunction. The role of mtDNA instability in retinal neuroinflammation, structural changes, and the role of the CGAS-STING pathway in these changes remains poorly understood.

Method: Proteomics and bulk RNA seq were performed on retinal tissue from 35 week old WT, polg mutator (Mut), sting knockout (KO), and a double knockout (DKO; polg mut+sting ko) mice. Retinal structure was evaluated by immunolabeling retinal flat-mounts for Iba1 (microglia), Brn3a (RGC) and IB4 (vasculature morphology), along with cryosectioning for H&E staining and tunel staining assessed cell death.

Results: Proteomics showed downregulation of oxidative phosphorylation (complex I) alongside upregulation of iron sulfur (fe-S) cluster assembly and heme biosynthesis proteins (Isca2, Nfu1, Fdx2, Fech, Abcb7) in Mut retinas indicating mitochondrial iron loading and compensatory biogenesis. Metabolic rewiring toward fatty acid beta-oxidation, amino acid catabolism as well as antioxidants, indicated mitochondrial stress. Anti-apoptotic protein down-regulation was accompanied by up-regulation of proapoptotic protein signatures in Mut retinas. This was further supported by tunel positive apoptotic cell death in Mut retinas as well as in DKO mic indicating sting independent apoptotic dysregulation. Low grade innate immune activation including Rigi pathway components was present in both Mut and DKO retinas. DKO vs Mut omics comparison revealed no significant differential regulation of innate immune signaling pathways implicating cGAS-STING independent rna sensing mechanisms. Morphologically Mut retinas displayed dystrophic and amoeboid microglia, vascular rarefaction and reduced retinal layer.

Conclusion: These findings suggests that mtDNA instability can drive a glaucoma phenotype encompassing metabolic and iron dysregulation, apoptotic cell death and microglial dystrophy through a cGAS-STING independent pathway. Rigi mediated rna sensing could be a candidate neuroinflammatory mechanism for understanding and potentially targeting IOP independent glaucoma progression.

